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Abstract Title

Internal Validation of an Integrated Deep Learning models for 3D Carotid Artery Reconstruction and Stenosis Detection using MRA

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Purpose

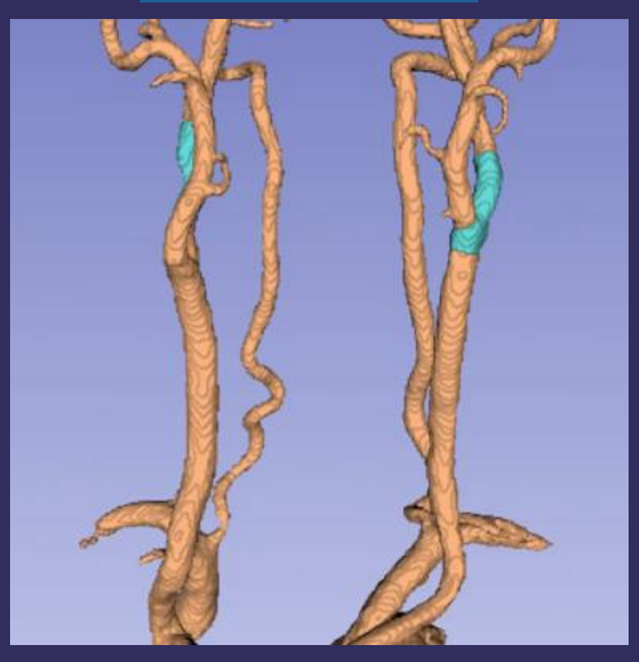
This study aimed to develop and evaluate the performance of an integrated deep learning model using TOF-MRA images for 3D carotid artery reconstruction (Vessel AI), stenosis and occlusion detection (Stenosis AI), and NASCET-based measurement (NASCET AI).

Architecture of AI

Vessel AI



Stenosis AI



NASCET AI



Carotid Vessel AI

Input data: Carotid TOF images, Analysis time: 4.5 sec
DICE score: 0.916

Carotid Stenosis AI

Input data: Carotid TOF images, Analysis time: 7.5 sec
Training data: 632 TOF-MRA studies

NASCET AI

NASCET AI measures the vessel diameter along the centerline.

Methods

Internal validation was performed using single-institution data.
A total of 450 MRA cases
- 154 stenosis patient cases, 296 normal cases
- 900 carotid arteries (176 stenotic arteries, 724 normal arteries)

Results

	Carotid artery- based analysis			Case-based Analysis		
	Stenosis AI	NASCET AI	Stenosis AI + NASCET AI	Stenosis AI	NASCET AI	Stenosis AI + NASCET AI
Sensitivity	71.6%	38.1%	84.7% (↑13%)	76.0%	46.8%	89.0% (↑13%)
Specificity	94.1%	92.0%	86.6% (↓7%)	88.5%	85.1%	76.0% (↓12%)
Accuracy	89.7%	81.4%	86.2%	84.2%	72.0%	80.4%
PPV	74.6%	53.6%	60.6%	77.5%	62.1%	65.9%
NPV	93.2%	85.9%	95.9%	87.6%	75.4%	93.0%

Each models had a high specificity, above ninety percent on an artery basis, and a high accuracy, but the sensitivity was lower than 80%.

For this reason we decided not to use them separately. We planned to combine the two models, accepting a loss of specificity in order to gain sensitivity.

As a result, on an artery basis the sensitivity rose to 85% percent.

Conclusion

Unlike cerebral aneurysms, carotid lesions lack clear demarcation, and the initial training dataset (600 cases) was relatively insufficient; however, the model showed meaningful performance. Future large-scale data collection and standardization of stenosis labeling will help overcome these limitations and further enhance the model's performance.

This abstract was also presented at the 2026 KJJC Conference in Osaka.